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Top Skills

Healthcare Information Technology (HIT)

AI for Business

Artificial Intelligence (AI)

Certifications

Matlab Onramp

Introduction to Career Skills in Data Analytics Introduction to Career Skills in Data Analytic

Essential Math for Machine Learning: Python Edition

Computational Approaches to Memory and Plasticity (CAMP) Summer School

Autodesk Inventor 2022 Essential Training

Honors-Awards

Presentation: 39th Indian Engineering Congress (Biomedical Science & Technology)

Oral Presentation: Leveraging Brain Channels for EEG Authentication Across Emotional States

Publications

Calophyllum oil prospective as alternative fuel for diesel engine.

Diagnostic Classification of ASD Improves with Structural Connectivity of DTI and Logistic Regression

Autism spectrum disorder diagnosis using fractal and non-fractal-based functional connectivity analysis and machine learning methods

Chetan Rakshe

PhD Researcher @ IIT (BHU) | Computational Neuroscience | Neuro-AI & EEG Foundation Models | Machine Learning | Deep Learning | MATLAB

Pune, Maharashtra, India

Summary

Pursuing a PhD in computational neuroscience, where I am focused on applying my skills and knowledge to solve complex problems in the field of neuroscience. I am particularly interested in interdisciplinary research, as I believe that combining diverse perspectives and approaches can lead to innovative solutions to complex problems. Skilled in machine learning, deep learning, MATLAB, python, fusion 360 and advance excel. I am eager to apply my skills and knowledge to make a positive impact on the healthcare industry and improve human health and performance.

Experience

Indian Institute of Technology (Banaras Hindu University), Varanasi

Junior Research Fellow
March 2022 - December 2022 (10 months)
Varanasi, Uttar Pradesh, India

As a junior research fellow worked on the SERB sponsored project titled "Development of Sparse Inverse Co-variance based functional brain connectivity Schemes for the assessment of Shared Autistic Traits in Autism and Typical Development," my responsibilities included designing and conducting experiments to investigate functional brain connectivity in individuals with autism and typical development. This involved collecting and analyzing functional magnetic resonance imaging (fMRI) data from participants, as well as developing algorithms and statistical models to identify patterns of connectivity that may be associated with shared autistic traits. To this end, I worked closely with Dr. Jac Fredo AR and other members of the research team to develop and implement innovative techniques for analyzing fMRI data, including sparse inverse covariance analysis and network-based statistics.

In addition to it, I had the opportunity to apply my knowledge of machine learning to the analysis of the data we collected. Specifically, I used machine learning algorithms and techniques to identify patterns in the functional brain

Age- and Severity-Specific Deep Learning Models for Autism Spectrum Disorder Classification Using Functional Connectivity Measures

VARIOUS BIODIESEL PRODUCTION METHODS - A REVIEW

connectivity data that may be indicative of shared autistic traits. This included applying techniques such as feature selection, dimensionality reduction, and classification to identify connectivity patterns that were unique to individuals with autism or typical development, and to explore whether these patterns were related to other measures of autism severity or cognitive function.
I enjoyed the opportunity to explore new approaches to data analysis and to see how machine learning could be used to uncover insights about brain function and behavior in individuals with autism.

Education

Indian Institute of Technology (Banaras Hindu University), Varanasi

Doctor of Philosophy - PhD, Biomedical/Medical Engineering · (December 2022 - December 2025)

Savitribai Phule Pune University

Bachelor of Engineering - BE, Mechanical Engineering · (2017 - 2021)

Fergusson College

HSC, Science · (2015 - 2017)